NYPD Shooting Data Project

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Setup

This dataset contains NYPD Shooting Incident Reports from January 2006 to December 2024 across precincts in New York City.

```
url_in <- "https://data.cityofnewyork.us/api/views/833y-fsy8/rows.csv"
```

```
nypd_data <- read_csv(url_in)
```

```
## Rows: 29744 Columns: 21
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (12): OCCUR_DATE, BORO, LOC_OF_OCCUR_DESC, LOC_CLASSFCTN_DESC, LOCATION...
## dbl   (5): INCIDENT_KEY, PRECINCT, JURISDICTION_CODE, Latitude, Longitude
## num   (2): X_COORD_CD, Y_COORD_CD
## lgl   (1): STATISTICAL_MURDER_FLAG
## time  (1): OCCUR_TIME
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# remove unnecessary columns and changed date column to date data type
nypd_data <- nypd_data %>%
  select(-c(JURISDICTION_CODE, X_COORD_CD, Y_COORD_CD)) %>%
  mutate(OCCUR_DATE = mdy(OCCUR_DATE))
# just the basic time and location information for incidents
incident_basic <- nypd_data %>% select(c(INCIDENT_KEY, OCCUR_DATE, OCCUR_TIME, BORO, Latitude, Longitud
```

```
summary(nypd_data)
```

```
##   INCIDENT_KEY         OCCUR_DATE          OCCUR_TIME           BORO
##  Min.   :  9953245   Min.   :2006-01-01   Length:29744      Length:29744
##  1st Qu.: 67321140   1st Qu.:2009-10-29   Class1:hms        Class :character
##  Median :109291972   Median :2014-03-25   Class2:difftime   Mode  :character
##  Mean   :133850951   Mean   :2014-10-31   Mode  :numeric
##  3rd Qu.:214741917   3rd Qu.:2020-06-29
##  Max.   :299462478   Max.   :2024-12-31
##
##  LOC_OF_OCCUR_DESC     PRECINCT      LOC_CLASSFCTN_DESC LOCATION_DESC
```

```
##  Length:29744       Min.   :  1.00   Length:29744       Length:29744
##  Class :character   1st Qu.: 44.00   Class :character   Class :character
##  Mode  :character   Median : 67.00   Mode  :character   Mode  :character
##                     Mean   : 65.23
##                     3rd Qu.: 81.00
##                     Max.   :123.00
##
##  STATISTICAL_MURDER_FLAG PERP_AGE_GROUP       PERP_SEX
##  Mode :logical           Length:29744       Length:29744
##  FALSE:23979             Class :character   Class :character
##  TRUE :5765              Mode  :character   Mode  :character
##
##
##
##
##   PERP_RACE          VIC_AGE_GROUP        VIC_SEX            VIC_RACE
##  Length:29744       Length:29744       Length:29744       Length:29744
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##
##     Latitude       Longitude        Lon_Lat
##  Min.   :40.51   Min.   :-74.25   Length:29744
##  1st Qu.:40.67   1st Qu.:-73.94   Class :character
##  Median :40.70   Median :-73.91   Mode  :character
##  Mean   :40.74   Mean   :-73.91
##  3rd Qu.:40.83   3rd Qu.:-73.88
##  Max.   :40.91   Max.   :-73.70
##  NA's   :97      NA's   :97
```

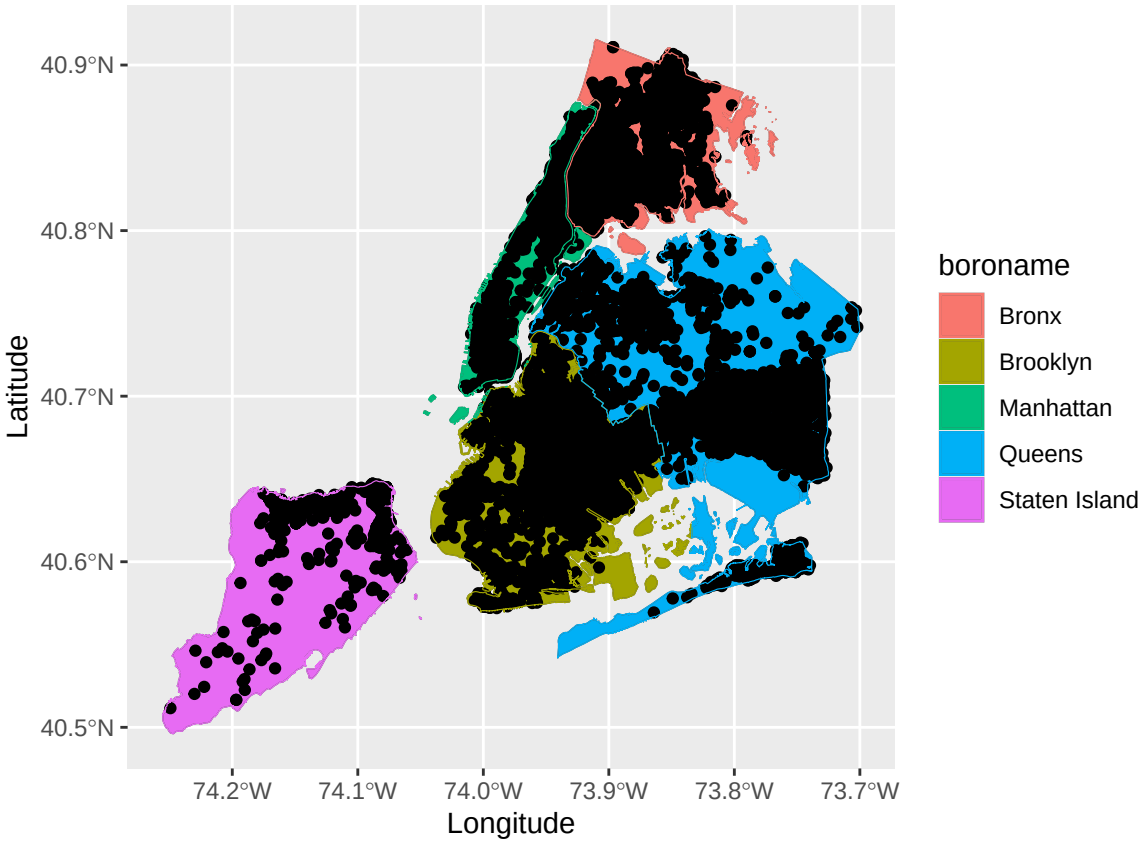
Analysis

First we will start with some general analysis.

Plot showing all incidents by location

```
nyc_map <-read_sf("https://data.cityofnewyork.us/resource/gthc-hcne.geojson")
incident_plot <- ggplot() + geom_sf(data=nyc_map, mapping=aes(fill=boroname)) + geom_point(data=(incide
incident_plot
```

```
## Warning: Removed 78 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Looking at the incident map, Staten Island has the fewest cases, mostly clustered along the northern and northeastern part of the island. Brooklyn, Manhattan, the Bronx, and east/central Queens have high case density. Additionally, northwest and southeast Queens have a few clusters with high case density. Due to the high case density covering most of NYC, several locations such as the Bronx Zoo, Central Park, and JFK airport can be easily picked out from the mass of surrounding incidents.

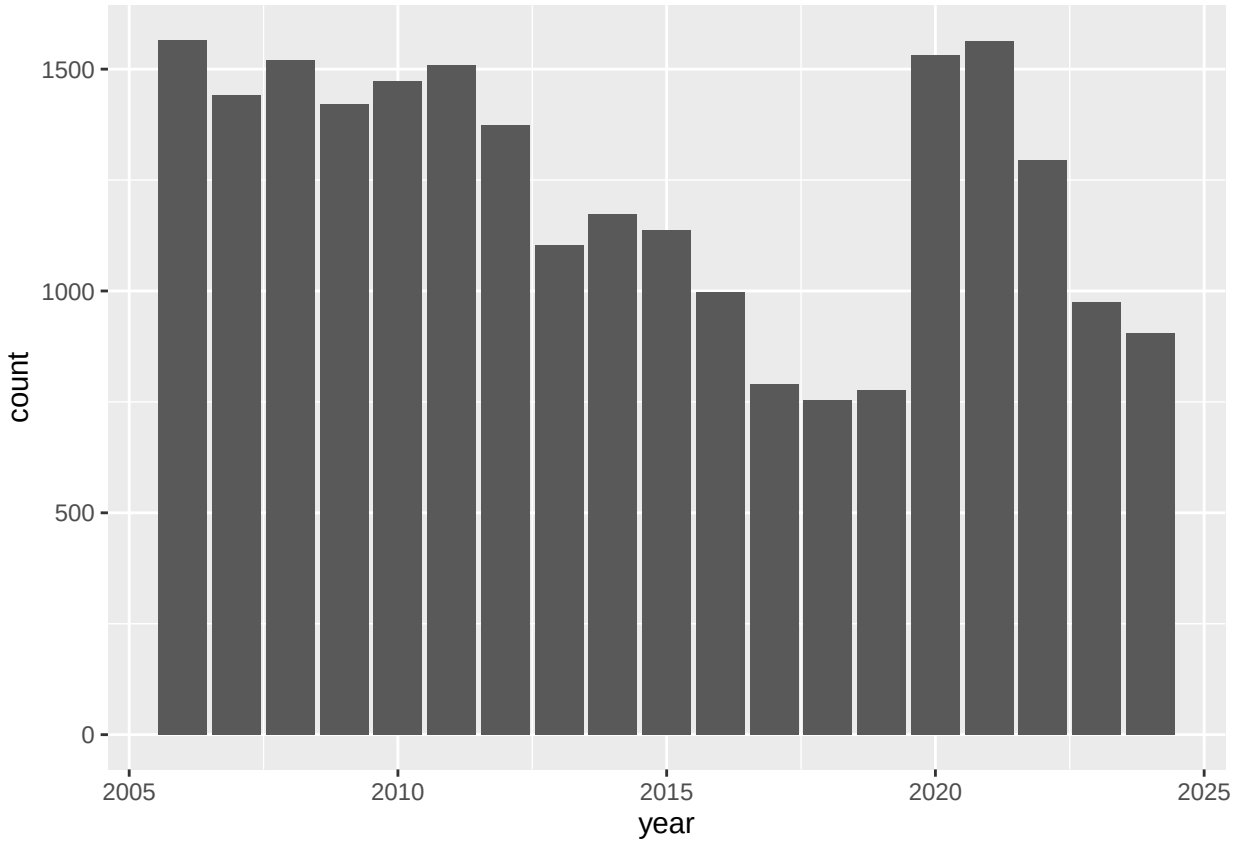
Total number of incidents each year

```
incidents_by_year <- incident_basic %>% unique.data.frame() %>% count(year(OCCUR_DATE))
incidents_by_year <- rename(incidents_by_year, year = `year(OCCUR_DATE)`, count = n)
incidents_by_year
```

```
## # A tibble: 19 x 2
##     year count
##    <dbl> <int>
##  1  2006  1566
##  2  2007  1442
##  3  2008  1520
##  4  2009  1420
##  5  2010  1473
##  6  2011  1509
##  7  2012  1373
##  8  2013  1103
##  9  2014  1172
## 10  2015  1138
## 11  2016   997
## 12  2017   789
```

```
## 13  2018   754
## 14  2019   777
## 15  2020  1531
## 16  2021  1562
## 17  2022  1294
## 18  2023   974
## 19  2024   904
```

```
ggplot(incidents_by_year, aes(year, count)) + geom_col()
```



The year with the most recorded incidents was 2006, with 2020 and 2021 close behind. 2018 had the fewest. Overall the annual number of incidents seemed to be trending downward until 2020 at which point it sharply increased.

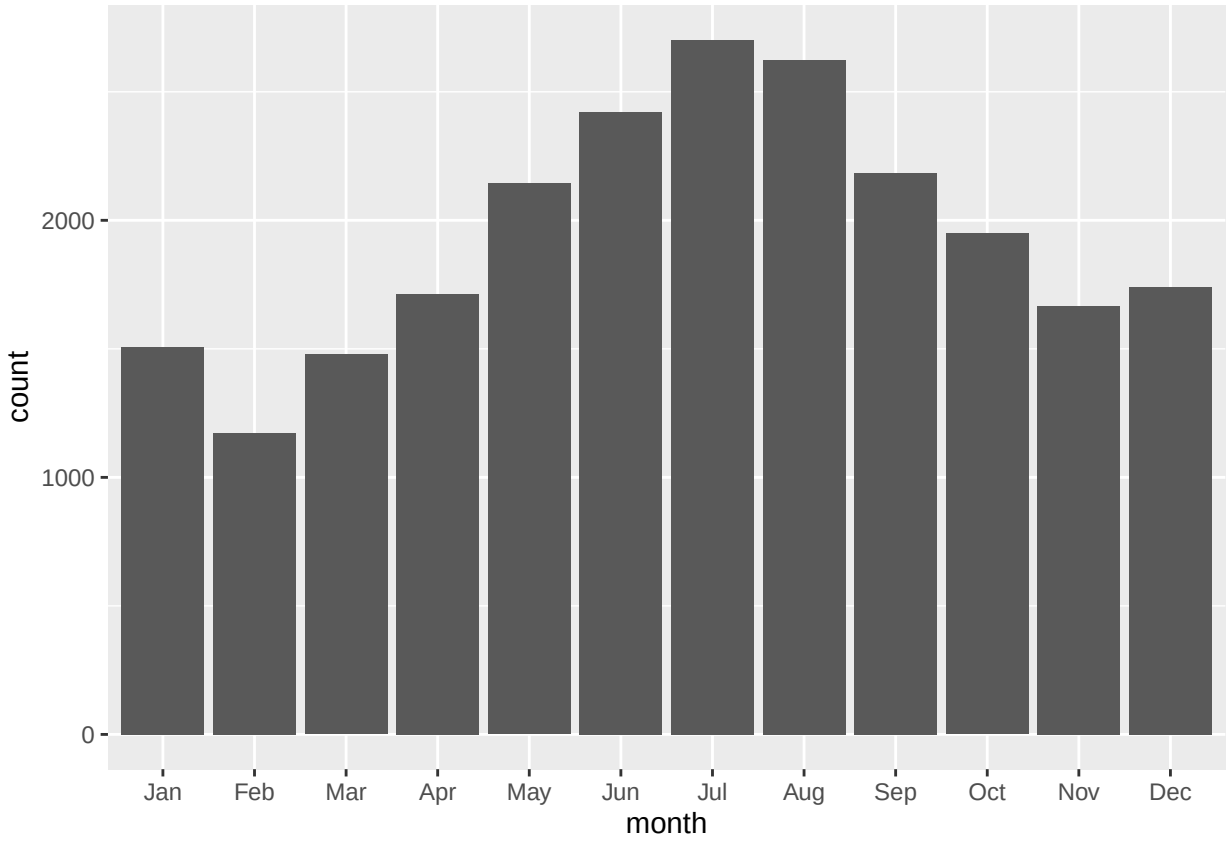
Total number of incidents combined by month

```
incidents_by_month <- incident_basic %>% unique.data.frame() %>% count(month(OCCUR_DATE, label = TRUE))
incidents_by_month <- rename(incidents_by_month, month = `month(OCCUR_DATE, label = TRUE)`, count = n)
incidents_by_month
```

```
## # A tibble: 12 x 2
##    month count
##    <ord> <int>
##  1 Jan    1508
##  2 Feb    1171
##  3 Mar    1478
```

```
##  4 Apr    1714
##  5 May    2143
##  6 Jun    2421
##  7 Jul    2703
##  8 Aug    2622
##  9 Sep    2183
## 10 Oct    1949
## 11 Nov    1666
## 12 Dec    1740
```

```
ggplot(incidents_by_month, aes(month, count)) + geom_col()
```



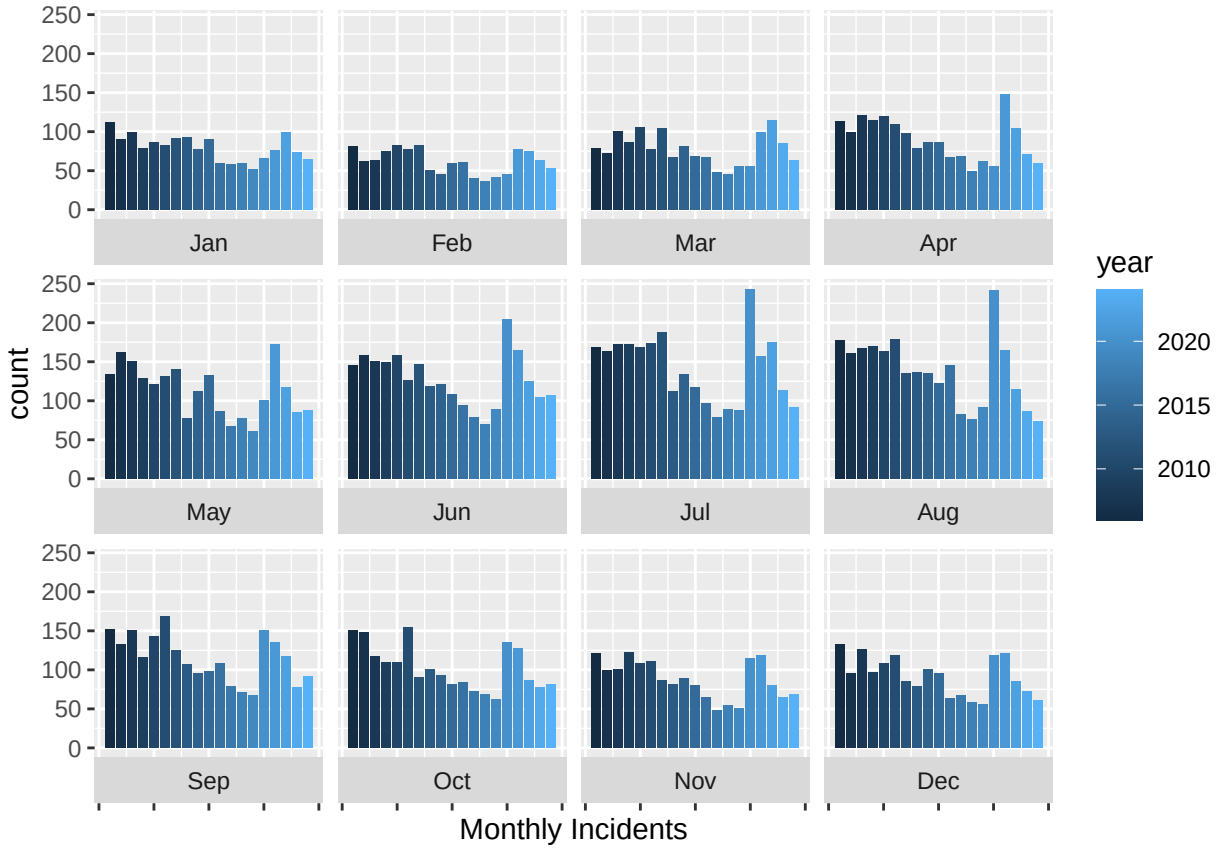
First I wanted to look at combined totals for each month over the 19 years to see what general pattern emerged. February is the month with the fewest combined number of incidents. July has the most. Then I looked at the months across each year.

Incidents by month

```
incidents_monthly <- incident_basic %>% unique.data.frame() %>% count(year(OCCUR_DATE), month(OCCUR_DATE
incidents_monthly <- rename(incidents_monthly, year = `year(OCCUR_DATE)`, month = `month(OCCUR_DATE, lab
# grouping by year
ggplot(incidents_monthly) + geom_col(aes(x=month,y=count,fill=month)) + labs(x='Monthly Incidents') + th
```




```
# grouping by month
ggplot(incidents_monthly) + geom_col(aes(x=year,y=count,fill=year)) + labs(x='Monthly Incidents') + ther
```



Days with the most incidents

```
incident_basic %>% unique.data.frame() %>% count(OCCUR_DATE) %>% arrange(desc(n))
```

```
## # A tibble: 6,431 x 2
##    OCCUR_DATE     n
##    <date>     <int>
##  1 2020-07-05    30
##  2 2011-09-04    21
##  3 2020-08-15    21
##  4 2020-06-27    17
##  5 2009-04-26    16
##  6 2014-07-06    16
##  7 2020-07-13    16
##  8 2020-08-22    16
##  9 2008-01-01    15
## 10 2008-07-26    15
## # i 6,421 more rows
```

```
most_incidents <- tibble(incident_basic %>% unique.data.frame() %>% count(date(OCCUR_DATE)))
most_incidents <- most_incidents %>% rename(date = `date(OCCUR_DATE)`, incidents = n) %>% slice_max(ord
knitr::kable(most_incidents, caption = "Days with the most incidents")
```

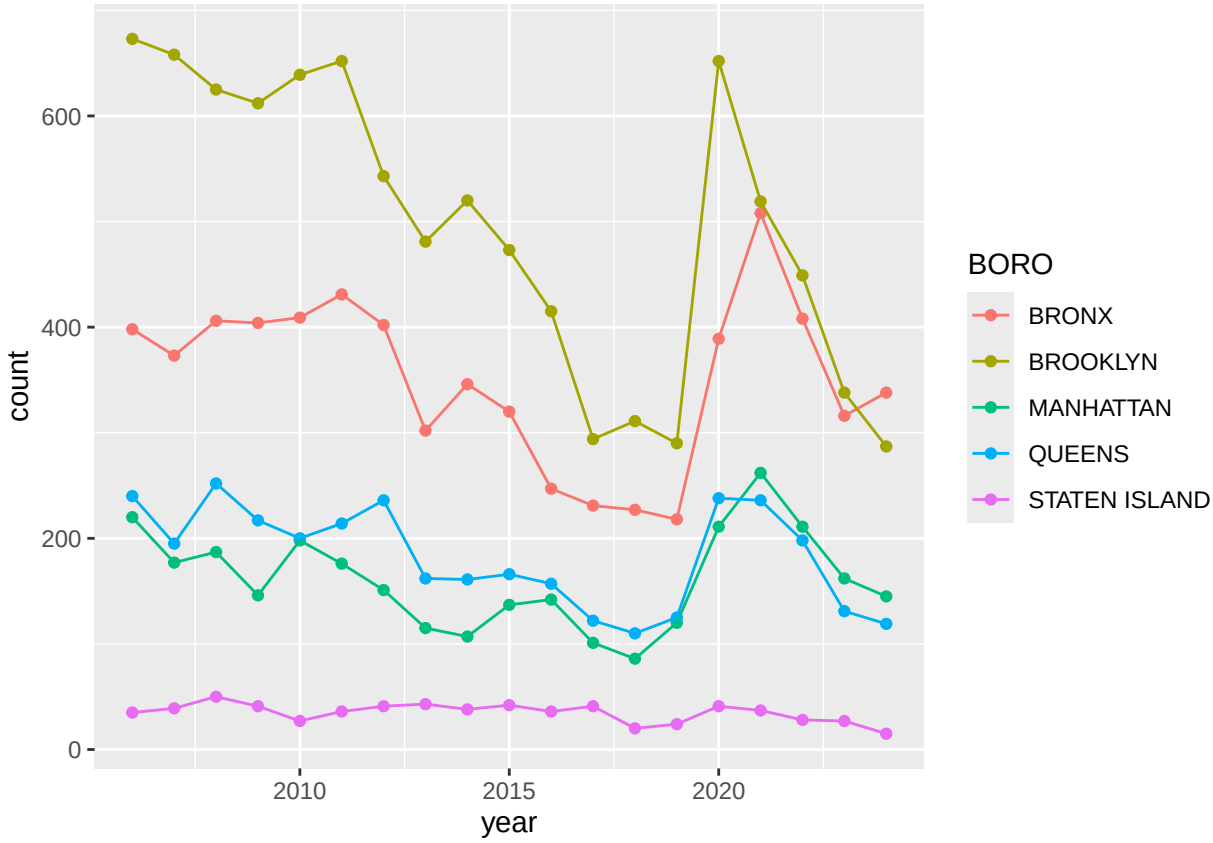
Table 1: Days with the most incidents

date	incidents
2020-07-05	30
2011-09-04	21
2020-08-15	21
2020-06-27	17
2009-04-26	16
2014-07-06	16
2020-07-13	16
2020-08-22	16

Now some more specific analyisis.

Incidents per year by borough

```
boro_per_year <- incident_basic %>% unique.data.frame() %>% group_by(BORO) %>% count(year(OCCUR_DATE))
boro_per_year <- rename(boro_per_year, year = `year(OCCUR_DATE)`, count = n)
ggplot(boro_per_year,aes(year, count, color = BORO))+geom_point() + geom_line()
```



As evident by the incident map earlier and this plot here, Brooklyn has the highest number of shootings almost every year while Staten Island has a very low overall yearly case load. Although differing in population density and size, Manhattan and Queens are pretty evenly matched in incident reports. Brooklyn incident reports surged in 2020 and then started decreasing again where the Bronx had a significant increase in 2020, peaked in 2021, and then started decreasing again until 2024.

Looking at boroughs and precincts by total

```r
incident_basic %>% unique.data.frame() %>% count(BORO)
```

```
## # A tibble: 5 x 2
##   BORO              n
##   <chr>         <int>
## 1 BRONX          6673
## 2 BROOKLYN       9431
## 3 MANHATTAN      3054
## 4 QUEENS         3479
## 5 STATEN ISLAND   661
```

```r
nypd_data %>% select(INCIDENT_KEY,OCCUR_DATE,OCCUR_TIME,BORO,PRECINCT) %>% unique.data.frame() %>% grou
```

```
## # A tibble: 77 x 3
## # Groups:   PRECINCT [77]
##    PRECINCT BORO         n
##       <dbl> <chr>    <int>
##  1       75 BROOKLYN  1336
##  2       73 BROOKLYN  1265
##  3       67 BROOKLYN  1065
##  4       79 BROOKLYN   851
##  5       44 BRONX      838
##  6       40 BRONX      815
##  7       47 BRONX      813
##  8       46 BRONX      758
##  9       77 BROOKLYN   719
## 10       42 BRONX      711
## # i 67 more rows
```

Incidents with a minor as the perpetrator

```r
incidents_perps <- incident_basic %>% unique.data.frame()
incidents_perps <- incidents_perps %>% left_join(nypd_data%>%select(INCIDENT_KEY,PERP_AGE_GROUP)%>%filt
```

```
## Joining with `by = join_by(INCIDENT_KEY)`
```

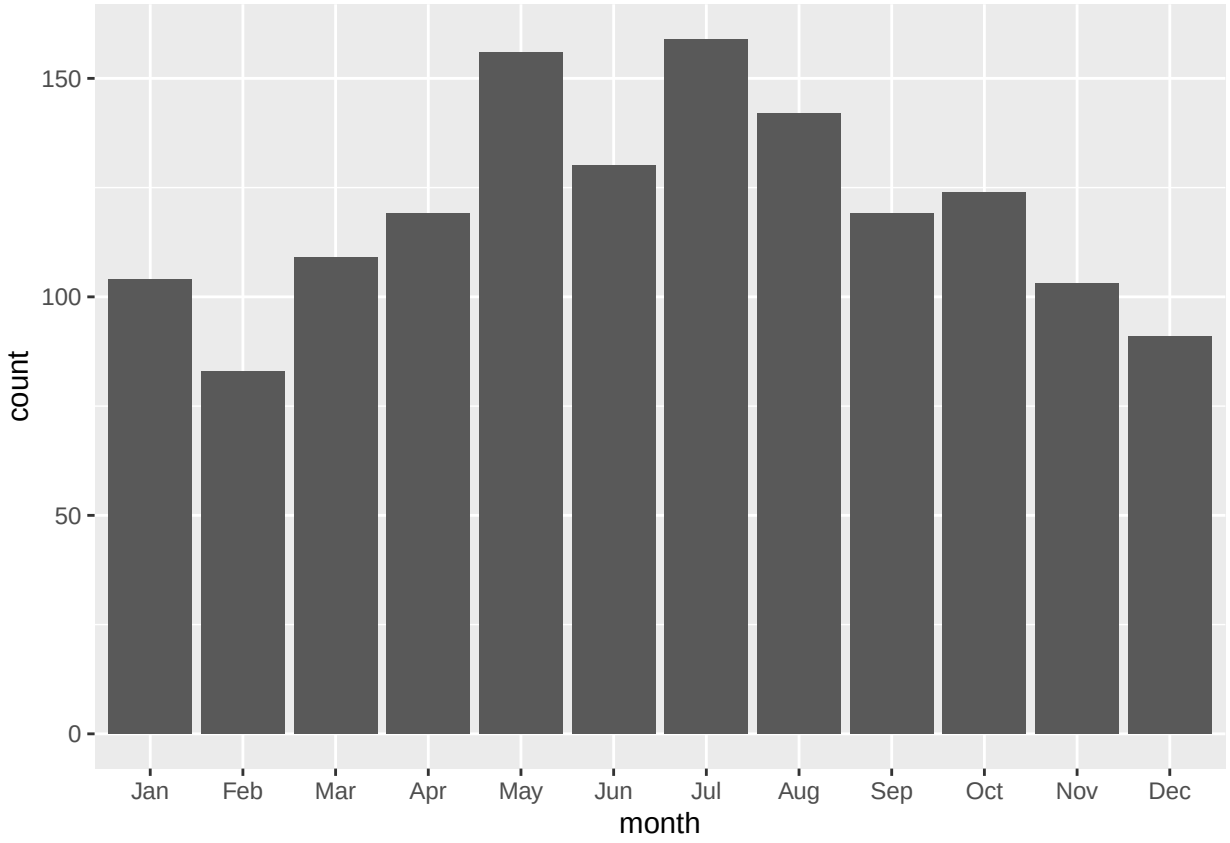
```r
incidents_perp_minor <- incidents_perps %>% filter(PERP_AGE_GROUP=="<18")

incidents_perp_minor_month <- incidents_perp_minor %>% unique.data.frame() %>% count(month(OCCUR_DATE, 
incidents_perp_minor_month <- rename(incidents_perp_minor_month, month = `month(OCCUR_DATE, label = TRU
incidents_perp_minor_month
```

```
## # A tibble: 12 x 2
##    month count
##    <ord> <int>
##  1 Jan     104
##  2 Feb      83
##  3 Mar     109
##  4 Apr     119
##  5 May     156
```

```
##  6 Jun     130
##  7 Jul     159
##  8 Aug     142
##  9 Sep     119
## 10 Oct     124
## 11 Nov     103
## 12 Dec      91
```

```
ggplot(incidents_perp_minor_month, aes(month, count)) + geom_col()
```




```
incidents_perp_adult <- anti_join(incidents_perps, incidents_perp_minor)
```

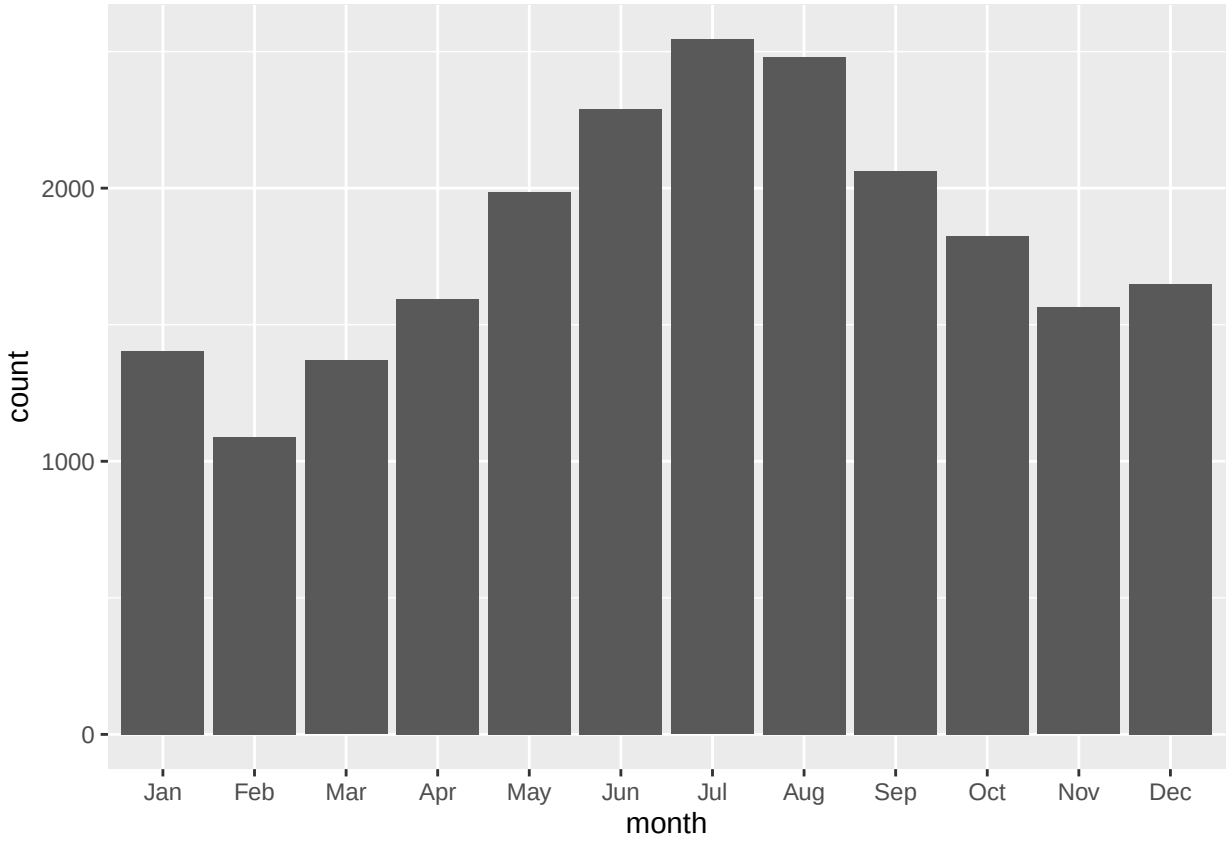
```
## Joining with `by = join_by(INCIDENT_KEY, OCCUR_DATE, OCCUR_TIME, BORO,
## Latitude, Longitude, PERP_AGE_GROUP)`
```

```
incidents_perp_adult_month <- incidents_perp_adult %>% unique.data.frame() %>% count(month(OCCUR_DATE, l
incidents_perp_adult_month <- rename(incidents_perp_adult_month, month = `month(OCCUR_DATE, label = TRU
incidents_perp_adult_month
```

```
## # A tibble: 12 x 2
##    month count
##    <ord> <int>
##  1 Jan    1404
##  2 Feb    1088
```

```
##  3 Mar    1369
##  4 Apr    1595
##  5 May    1987
##  6 Jun    2291
##  7 Jul    2544
##  8 Aug    2480
##  9 Sep    2064
## 10 Oct    1825
## 11 Nov    1563
## 12 Dec    1649
```

```
ggplot(incidents_perp_adult_month, aes(month, count)) + geom_col()
```



I was curious if there would be a noticeable difference in the overall trends between shootings involving minor and adult perpetrators. My assumption was that the trend would show higher totals in the summer months since that's when most kids are out of school. It is interesting that while February is still the lowest and July the highest, May is quite a bit higher and June lower than expected from the adult/general trend. The under 18 set is signifcantly smaller than the adult set so the numbers themselves are lower but the overall trend is a lot less curved.

##Just out of curiosity

Incidents happening at the same time

```
incident_basic %>% unique.data.frame() %>% count(OCCUR_DATE, OCCUR_TIME) %>% arrange(desc(n))
```

```
## # A tibble: 23,183 x 3
```

```
##    OCCUR_DATE OCCUR_TIME     n
##    <date>     <time>     <int>
##  1 2022-09-19 20:00          3
##  2 2006-05-23 15:46          2
##  3 2006-06-22 19:14          2
##  4 2006-07-16 04:30          2
##  5 2006-08-06 00:45          2
##  6 2006-08-13 19:00          2
##  7 2006-09-11 18:22          2
##  8 2006-10-07 21:50          2
##  9 2006-12-23 01:30          2
## 10 2007-02-17 04:00          2
## # i 23,173 more rows
```

Incidents with multiple perpetrators or victims

```
incident_basic %>% count(OCCUR_DATE,OCCUR_TIME, INCIDENT_KEY) %>% arrange(desc(n))
```

```
## # A tibble: 23,298 x 4
##    OCCUR_DATE OCCUR_TIME INCIDENT_KEY     n
##    <date>     <time>            <dbl> <int>
##  1 2024-02-12 16:32         282117921    20
##  2 2018-01-06 21:05         173354054    18
##  3 2023-02-10 14:29         263503175    16
##  4 2006-08-07 22:06          23749375    12
##  5 2006-10-09 19:41          24717013    12
##  6 2007-07-30 22:20          33478089    12
##  7 2007-08-11 00:09          33706902    12
##  8 2007-11-18 01:00          35803777    12
##  9 2009-09-22 15:22          66027258    12
## 10 2010-04-12 23:59          72195829    12
## # i 23,288 more rows
```

```
nypd_data %>% filter(INCIDENT_KEY==282117921)
```

```
## # A tibble: 20 x 18
##    INCIDENT_KEY OCCUR_DATE OCCUR_TIME BORO  LOC_OF_OCCUR_DESC PRECINCT
##           <dbl> <date>     <time>     <chr> <chr>                <dbl>
##  1    282117921 2024-02-12 16:32      BRONX INSIDE                  44
##  2    282117921 2024-02-12 16:32      BRONX INSIDE                  44
##  3    282117921 2024-02-12 16:32      BRONX INSIDE                  44
##  4    282117921 2024-02-12 16:32      BRONX INSIDE                  44
##  5    282117921 2024-02-12 16:32      BRONX INSIDE                  44
##  6    282117921 2024-02-12 16:32      BRONX INSIDE                  44
##  7    282117921 2024-02-12 16:32      BRONX INSIDE                  44
##  8    282117921 2024-02-12 16:32      BRONX INSIDE                  44
##  9    282117921 2024-02-12 16:32      BRONX INSIDE                  44
## 10    282117921 2024-02-12 16:32      BRONX INSIDE                  44
## 11    282117921 2024-02-12 16:32      BRONX INSIDE                  44
## 12    282117921 2024-02-12 16:32      BRONX INSIDE                  44
## 13    282117921 2024-02-12 16:32      BRONX INSIDE                  44
## 14    282117921 2024-02-12 16:32      BRONX INSIDE                  44
```

```
## 15    282117921 2024-02-12 16:32      BRONX INSIDE                  44
## 16    282117921 2024-02-12 16:32      BRONX INSIDE                  44
## 17    282117921 2024-02-12 16:32      BRONX INSIDE                  44
## 18    282117921 2024-02-12 16:32      BRONX INSIDE                  44
## 19    282117921 2024-02-12 16:32      BRONX INSIDE                  44
## 20    282117921 2024-02-12 16:32      BRONX INSIDE                  44
## # i 12 more variables: LOC_CLASSFCTN_DESC <chr>, LOCATION_DESC <chr>,
## #   STATISTICAL_MURDER_FLAG <lgl>, PERP_AGE_GROUP <chr>, PERP_SEX <chr>,
## #   PERP_RACE <chr>, VIC_AGE_GROUP <chr>, VIC_SEX <chr>, VIC_RACE <chr>,
## #   Latitude <dbl>, Longitude <dbl>, Lon_Lat <chr>
```

```
nypd_data %>% filter(INCIDENT_KEY==16814010)
```

```
## # A tibble: 6 x 18
##   INCIDENT_KEY OCCUR_DATE OCCUR_TIME BORO  LOC_OF_OCCUR_DESC PRECINCT
##          <dbl> <date>     <time>     <chr> <chr>                <dbl>
## 1     16814010 2006-06-12 20:45      BRONX <NA>                    52
## 2     16814010 2006-06-12 20:45      BRONX <NA>                    52
## 3     16814010 2006-06-12 20:45      BRONX <NA>                    52
## 4     16814010 2006-06-12 20:45      BRONX <NA>                    52
## 5     16814010 2006-06-12 20:45      BRONX <NA>                    52
## 6     16814010 2006-06-12 20:45      BRONX <NA>                    52
## # i 12 more variables: LOC_CLASSFCTN_DESC <chr>, LOCATION_DESC <chr>,
## #   STATISTICAL_MURDER_FLAG <lgl>, PERP_AGE_GROUP <chr>, PERP_SEX <chr>,
## #   PERP_RACE <chr>, VIC_AGE_GROUP <chr>, VIC_SEX <chr>, VIC_RACE <chr>,
## #   Latitude <dbl>, Longitude <dbl>, Lon_Lat <chr>
```

This is where analysis gets tricky. These multi record incidents aren't actually as clear cut as they appear to be. Take the first incident listed here with 20 records. At first glance this appears to be a large mass shooting with half of the records being fatalities. But looking at news reports of the case, it turns out there was only 1 fatality and 5 wounded. How do we get to 20 records from 6 victims? It could be a function of the collection / reporting method or an attempt to anonymize public records but either way, it seems incidents with multiple perpetrators or victims involved, particularly if there was a fatality, each combination of perp/vic/flag is listed as a separate record for that incident. So in this case, there were 2 perpetrators, 5 victims (note: only 5 of the 6 are actually recorded here), with 1 being a fatality (True/False murder flag) which makes 20 combinations. The second incident I looked at shows this a little more clearly since there is no information recorded on the perpetrator. In this incident there were 3 victims - <18 M, 18-24 F, and 25-44 M - 1 was a fatality so each victim has a record with a true flag and a record with a false flag. This doesn't necessarily mean we can't trust the data, just that we have to be more careful with how we analyze it.

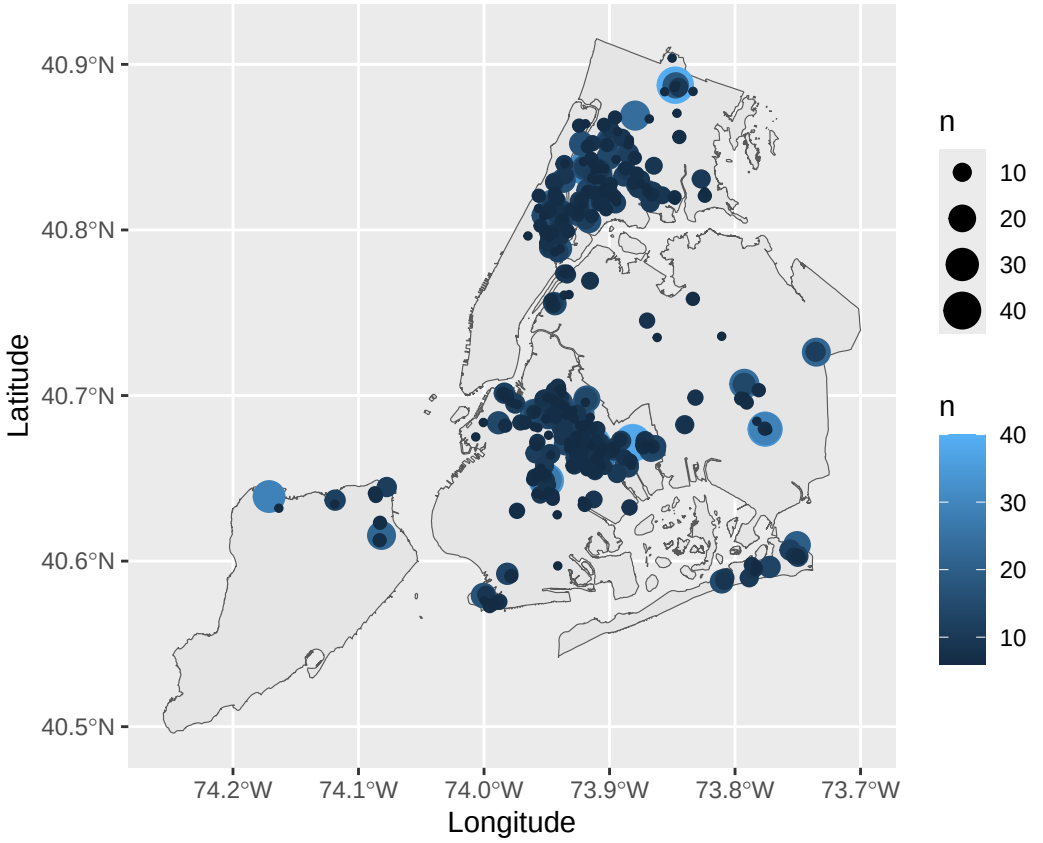
Locations where multiple incidents have been reported

```
counts <-incident_basic %>% unique.data.frame() %>% drop_na() %>% count(Latitude, Longitude) %>% arrang
counts
```

```
## # A tibble: 14,057 x 3
##    Latitude Longitude     n
##       <dbl>     <dbl> <int>
##  1     40.7     -73.9    40
##  2     40.9     -73.8    39
##  3     40.7     -73.9    38
##  4     40.7     -73.9    37
```

```
##  5     40.6     -74.0    34
##  6     40.7     -73.8    34
##  7     40.8     -73.9    31
##  8     40.6     -74.2    29
##  9     40.7     -73.8    29
## 10     40.6     -74.0    27
## # i 14,047 more rows
```

```
ggplot()+ geom_sf(data=nyc_map) + geom_point(data=(counts%>%filter(n>5)), mapping=aes(Longitude, Latitu
```



I was curious if there were any locations that have had multiple incidents over the years. On looking at a map of where the actual locations were for several of the high count spots, a number of these actually pointed to police precincts around the city. While it is possible that NYPD Precinct 75 has had 40 shootings outside and 38 inside, Precinct 47 has had 39 shootings, Precinct 73 has had 37, and so on, a quick glance at the (sparse) location descriptions make it more likely that the location of many of these incidents was put as the precinct possibly because the actual location was unknown/unreported/unverified or because that is where the incident report was filled out. Not all of these are locations of police precincts. For example, the location for the first record with 29 shootings leads to an apartment complex in Staten Island which is in agreement with the records listed in the dataset for those coordinates.

Bias

It is hard to avoid putting your own assumptions, experiences, and interpretations into the data analysis. To try and mitigate bias as much as possible, I avoided using any demographic information associated with location and perpetrator/victim profiles. I used general analysis for most of the report and restricted the use of analysis of perpetrator information to age <18 for the minor vs adult monthly comparison.